System Requirements Specification

Basic Cyber-Physical System Attack and Defense Testbed

Document	Project team	Revision / date
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3. Specific Requirements

This section explains what the testbed must do and what users should be able to observe. The intended users are industry professionals, cybersecurity researchers, and students studying industrial control system security. The project is being developed across two semesters. The current phase builds the PLC network, historian, sensors, and attack tests; the later phase can expand the physical process and web features as those designs are finalized.

3.0 Key Terms

Term	Meaning
PLC	Programmable Logic Controller: an industrial computer that reads sensors and controls equipment.
Attack computer	A computer connected to the test network that acts like an attacker already inside the industrial network.
Historian	The system that receives, displays, and stores PLC values, sensor readings, attack events, and test results.
PLC profile database	Stored information about each PLC, including its brand, model, protocol, network settings, and connected sensors.
Python test script	A reusable Python program that performs a normal-operation test or a controlled attack test.
Test run number	A unique number, such as Test Run 001, used to match one test with its historian records, network capture, and result.

3.1 Interface Requirements

These requirements describe how the PLCs, sensors, attack computer, historian, database, and monitoring tools exchange information.

Interface	Information entering the system	Expected result
PLCs and network	PLC values and messages sent through EtherNet/IP, PROFINET, or Modbus TCP	The historian receives and identifies data from the correct PLC.

Interface	Information entering the system	Expected result
Sensors	Digital or analog sensor readings connected to the PLCs	The historian shows whether the sensor is connected and displays its current reading.
Attack computer	Controlled internal-attack traffic or commands	The targeted PLC/network response is observed and recorded.
Historian and database	PLC data, sensor data, attack events, and stored PLC profiles	Users can view current activity and review saved test information.
Python test scripts	A selected script and target PLC	The test runs and reports whether it completed, stopped, or failed.
Network monitor	Traffic moving across the test network	Wireshark-compatible PCAP or PCAPNG evidence is produced.

IR-001 The system shall connect the available Allen-Bradley, Siemens, and Schneider Electric/Modicon PLCs through an Ethernet switch.

IR-002 The system shall communicate with the Allen-Bradley PLC using EtherNet/IP, the Siemens PLC using PROFINET, and the Schneider/Modicon PLC using Modbus TCP.

IR-003 For each PLC, the historian shall show whether at least one connected sensor is working and shall display the sensor's current value or on/off state.

IR-004 The attack computer shall be able to send controlled internal-attack traffic to a selected PLC on the test network.

IR-005 The historian shall receive, display, and store PLC states, sensor readings, communication problems, attack events, and test results.

IR-006 The PLC profile database shall store each PLC's brand, model, protocol, network settings, historian tags, input/output connections, and sensor information.

IR-007 At the end of a test, the system shall show Completed, Stopped, or Failed and shall include the test run number so the result can be matched to the correct records.

3.2 Functional Requirements

These requirements describe the main actions the completed testbed must perform.

3.2.1 Setup and Normal Operation

FR-001 The system shall demonstrate basic communication from each PLC, through the Ethernet switch, to the historian.

FR-002 When a supported PLC is selected, the system shall use its stored database profile to identify the PLC and load the information needed to communicate with it.

FR-003 Before an attack test begins, the historian shall record the normal PLC state and sensor reading so they can be compared with the values observed during the attack.

3.2.2 Testing and Monitoring

FR-004 The team shall be able to use reusable Python scripts to test PLC communication, sensor operation, logging, and the selected attack methods.

FR-005 The historian and network-monitoring tools shall begin recording before an attack starts and shall continue until the final test result is recorded.

FR-006 The system shall demonstrate and record one internal attack launched from the attack computer.

FR-007 The system shall demonstrate and record one external attack launched from outside the normal PLC network connection path.

FR-008 The system shall demonstrate and record one hardware attack in which an ESP32-controlled interception module interrupts or changes a sensor signal before it reaches the PLC.

FR-009 Before each test, the system shall assign a unique test run number, such as Test Run 001. The same number shall appear in the historian records, network-capture filename, and test result so all information from that test can be matched.

FR-010 The same Python test script shall be reusable on supported PLCs, with only the PLC-specific settings changed when necessary.

FR-011 After a test, the team shall be able to return the PLC and sensor to normal operation and confirm the normal values through the historian.

FR-012 If communication with a PLC is lost, the historian shall identify the disconnected PLC, stop showing its old value as current, and record when the problem occurred.

3.3 Performance Requirements

PER-001 The testbed shall support all three PLCs connected to the Ethernet switch at the same time.

PER-002 During normal communication, the historian shall display a changed PLC or sensor value within 2 seconds after receiving the change.

PER-003 The historian shall save each received PLC change, sensor change, attack event, or communication problem within 5 seconds.

PER-004 The time shown in historian records and network captures for the same event shall differ by no more than 1 second, allowing the records to be compared accurately.

3.4 Reliability and Availability Requirements

REL-001 A saved Python test script shall run at least three times under the same setup without requiring the test steps to be rewritten.

REL-002 If one PLC loses communication, records already collected from that PLC and the other PLCs shall remain available.

AVL-001 With the hardware powered and the setup instructions available, the team shall be able to start the historian, database, and monitoring tools and confirm that the testbed is ready within 10 minutes.

3.5 Security Requirements

SEC-001 The testbed shall operate on an isolated laboratory network so attack traffic does not reach the public Internet or the university's production network.

SEC-002 The testbed interface shall require an authorized username and password before a user can start an attack test or change PLC settings.

SEC-003 The system shall record who started each attack test, which Python script was used, which PLC was targeted, and when the test occurred.

SEC-004 Passwords and PLC login information shall not be written directly inside the Python scripts or included in test logs shown to users.

SEC-005 Python attack scripts shall only target the laboratory PLCs and the PLC inputs, outputs, or settings listed in the approved test description.

3.6 Maintainability Requirements

MNT-001 For each supported PLC, the project documentation shall explain the network setup, protocol setup, historian connection, input/output mapping, sensor connections, and setup/testing steps.

MNT-002 A new PLC profile shall be addable to the database without rewriting the main historian or testbed interface. This provides the project's plug-and-play support.

MNT-003 When a test cannot continue, the system shall show a simple error message that identifies the affected PLC or tool and explains what the user should check.

3.7 Portability Requirements

PRT-001 The project services that do not depend on a specific PLC vendor shall be able to run through Docker on the EGSMTPC mini computer or another compatible x86-64 computer.

PRT-002 The testbed's browser-based interface shall work in current desktop versions of Google Chrome and Microsoft Edge without requiring a browser extension.

PRT-003 PLC addresses, network ports, and file-save locations shall be changeable in a settings file without editing the Python program itself.

3.8 Milestone Traceability

Project work	Related requirements	Simple acceptance evidence
Milestone 1: PLC setup and basic communication	IR-001-IR-006; FR-001-FR-004	Each PLC communicates through the switch; its sensor value appears in the historian; its database profile is complete.
Milestone 2: sensors and internal attack	IR-003-IR-005; FR-003-FR-006; PER-001-PER-004	The internal attack runs from the attack computer, and the historian and network capture show what happened.
Milestone 3: external and ESP32 hardware attacks	FR-007-FR-012; REL-001-REL-002	Both attacks run, the results are recorded under a test run number, and normal operation is restored afterward.
System access, maintenance, and reuse	SEC-001-SEC-005; MNT-001-MNT-003; PRT-001-PRT-003	Login works; tests stay on the lab network; PLC documentation and profiles are complete; settings can be changed without rewriting the program.

A requirement is accepted when the stated behavior can be observed and confirmed by a matching test. If the client changes a required feature, timing value, or supported device, this document and the Test Plan will be updated together.